

The Main Quiz (10% of total grades)

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✓ Question 1

✓ 0.9/1 pt ↻ 3

Under the operations inherited from $\mathbb{R}^3$, does the set $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid -5x - 4y - z = 0 \right\}$ form a vector space?

☒ Yes

☐ No



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● Question 2

✓ 0/1 pt ↻ 5

Under the usual matrix operations, does the set $\left\{ \begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\}$ form a vector space?

☐ Yes

☐ No

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● Question 3

✓ 0/1 pt ↻ 5

Given a set of vectors V , with $\vec{u}, \vec{v}, \vec{w} \in V$ and $a, b, c \in \mathbb{R}$, which of the following are NOT required for a vector space? Check all that apply.

☐ Closure of vector addition: $\vec{u} + \vec{v} \in V$

☐ $(a + b)\vec{v} = a\vec{v} + b\vec{v}$

☐ Closure of vector multiplication: $\vec{u}\vec{v} \in V$

☐ Closure under scalar multiplication: $a\vec{u} \in V$

☐ Commutativity of vector multiplication: $\vec{u}\vec{v} = \vec{v}\vec{u}$

☐ Commutativity of vector addition: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

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● Question 4

✓ 0/1 pt ↻ 5

Select all of the following that are vector spaces

- ☐ The set of all continuous functions defined on the closed interval $[1,3]$ with the standard operations
- ☐ The set of real numbers with the standard operations
- ☐ The set of all 3×4 matrices with the standard operations
- ☐ The set of all integers with the standard operations
- ☐ The set of all polynomials of degree exactly 4 with the standard operations

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● Question 5

✓ 0/1 pt ↻ 5

Is $\begin{bmatrix} -56 \\ -36 \\ 6 \end{bmatrix}$ in the spanning set of $\begin{bmatrix} 3 \\ -5 \\ -2 \end{bmatrix}, \begin{bmatrix} 6 \\ -8 \\ 8 \end{bmatrix}, \begin{bmatrix} 6 \\ -2 \\ 44 \end{bmatrix}$?

Select an answer ▼

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● Question 6

✓ 0/1 pt ↻ 5

Let $\mathbf{v}_1 = \begin{bmatrix} -8 \\ -3 \\ -2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 9 \\ 3 \\ 4 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} -11 \\ -3 \\ -8 \end{bmatrix}$.

Do these vectors span $\mathbb{R}^3$?

- ☐ No, $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ do not span $\mathbb{R}^3$
- ☐ Yes, $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ do span $\mathbb{R}^3$

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● Question 7

✓ 0/1 pt ↻ 5

True or False: The vectors $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ span $\mathbb{R}^3$.

- ☐ False
- ☐ True

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● Question 8

✓ 0/1 pt ↻ 5

Which of the following are spanning sets of $\mathbb{R}^3$? (These should be easy to do by just looking at them.)

$\{(1, 0, 0), (0, 5, 0), (7, 5, -9), (0, 0, 6)\}$

☐ No this set does NOT span $\mathbb{R}^3$

☐ Yes this set spans $\mathbb{R}^3$

$\{(8, 6, 7), (16, 12, 14), (3, 2, 1)\}$

☐ Yes this set spans $\mathbb{R}^3$

☐ No this set does NOT span $\mathbb{R}^3$

$\{(11, 1, 6), (2, 3, 9)\}$

☐ No this set does NOT span $\mathbb{R}^3$

☐ Yes this set spans $\mathbb{R}^3$

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● Question 9

✓ 0/1 pt ↻ 5

Given that $V = \{v_1 = [7 \ -3 \ -5], v_2 = [5 \ 1 \ 5], v_3 = [26 \ -8 \ -10], v_4 = [1 \ 2 \ m]\}$. Find m such that V is a spanning set of $\mathbb{R}^3$.

Answer: m _____

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● Question 10

✓ 0/1 pt ↻ 5

Is the vector $\begin{bmatrix} 12 \\ -26 \\ -16 \end{bmatrix}$ in the set generated by (spanned by) the set $\left\{ \begin{bmatrix} 3 \\ -5 \\ -4 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} \right\}$

If so, find and enter the necessary coefficients. If not, enter DNE in both boxes.

$$\begin{bmatrix} 12 \\ -26 \\ -16 \end{bmatrix} = \boxed{} \begin{bmatrix} 3 \\ -5 \\ -4 \end{bmatrix} + \boxed{} \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$$

● Question 11

✔ 0/1 pt ↻ 5

Can we make the polynomial $h(x) = 34x^2 + 27x + 6$ by a linear combination of the following polynomials?

$$f_1(x) = 6x^2 + 3x + 1$$

$$f_2(x) = 11x^2 + 9x + 2$$

$$f_3(x) = -6x^2 - 4x - 1$$

Answer:

- ☐ Yes, this is possible.
- ☐ No, this is not possible.

If so, fill in the blanks to make the statement true. [If no solutions are possible, enter DNE into each blank.]

$$h(x) = \underline{\hspace{2cm}} f_1(x) + \underline{\hspace{2cm}} f_2(x) + \underline{\hspace{2cm}} f_3(x)$$

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● Question 12

✔ 0/1 pt ↻ 5

Can we make the matrix $H = \begin{bmatrix} -3 & -7 \\ -23 & -9 \end{bmatrix}$ by a linear combination of the following matrices?

$$A_1 = \begin{bmatrix} 1 & 1 \\ 6 & 2 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 0 & 3 \\ 3 & 2 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} 0 & -2 \\ 2 & 1 \end{bmatrix}$$

Answer:

- ☐ Yes, this is possible.
☐ No, this is not possible.

If so, fill in the blanks to make the statement true. [If no solutions are possible, enter DNE into each blank.]

$$H = \underline{\hspace{1cm}} A_1 + \underline{\hspace{1cm}} A_2 + \underline{\hspace{1cm}} A_3 + \underline{\hspace{1cm}} A_4$$

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● Question 13

✓ 0/1 pt ↻ 5

Write $\mathbf{y}$ as the sum of two vectors, $\mathbf{w}_1 + \mathbf{w}_2$, where $\mathbf{w}_1$ is in $\text{Span}(\mathbf{u})$.

$$\mathbf{y} = \begin{bmatrix} -11 \\ 10 \end{bmatrix} \text{ and } \mathbf{u} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\mathbf{w}_1 = \boxed{\hspace{3cm}}$$

$$\mathbf{w}_2 = \boxed{\hspace{3cm}}$$

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● Question 14

✓ 0/1 pt ↻ 5

Assume $\vec{u}$ and $\vec{v}$ are vectors, and suppose $\vec{w}$ is in $\text{span}\{\vec{u}, \vec{v}\}$.

This means: (select all that apply)

- ☐ $\vec{w}$ is a linear combination of $\vec{u}$ and $\vec{v}$.
- ☐ There exist scalars a and b such that $\vec{w} = a\vec{u} + b\vec{v}$.
- ☐ $\vec{w} = \vec{u}$
- ☐ $\vec{w} = \vec{v}$
- ☐ $\vec{w} = \vec{u} + \vec{v}$
- ☐ None of the above.

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● Question 15

✓ 0/1 pt ↻ 5

The *span* of a set of vectors is the:

- ☐ set of all solutions of vector equations in that set
- ☐ set of all graphs of the vectors in that set
- ☐ set of all linear combinations of vectors in that set
- ☐ set of all scalar weights of vectors in that set

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● Question 16

✓ 0/1 pt ↻ 5

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are three nonzero vectors in $\mathbb{R}^3$ such that no two are parallel, then $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is all of $\mathbb{R}^3$.

- ☐ True
- ☐ False

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● Question 17

✓ 0/1 pt ↻ 5

The span of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is a parallelogram with vertices $(0, 0)$, $(1, 0)$, $(2, 1)$, and $(1, 1)$.

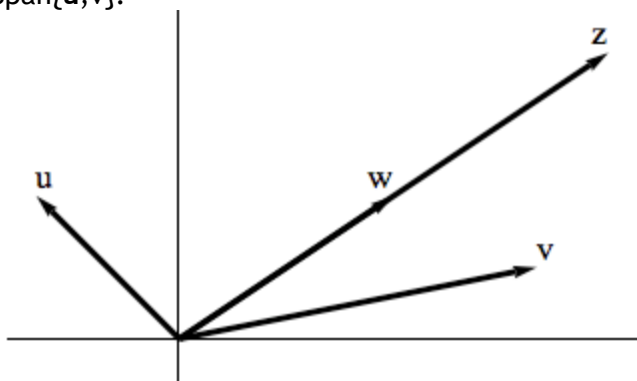
- ☐ True
- ☐ False

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● Question 18

✓ 0/1 pt ↻ 5

Consider the figure below in $\mathbb{R}^2$. Doing NO calculations, answer the following: Is w contained in $\text{Span}\{u, v\}$?



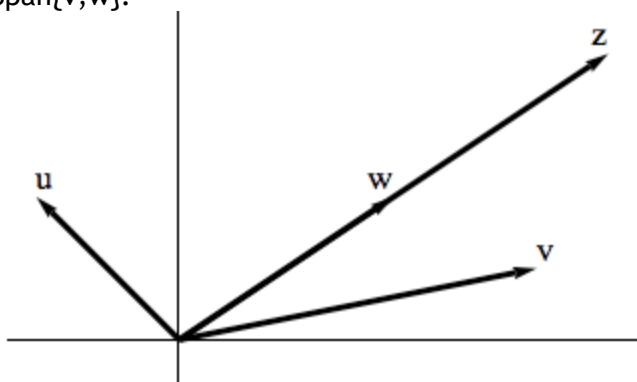
- ☐ No
☐ Yes

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● Question 19

✓ 0/1 pt ↻ 5

Consider the figure below in $\mathbb{R}^2$. Doing NO calculations, answer the following: Is u contained in $\text{Span}\{v, w\}$?



- ☐ Yes
☐ No

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● Question 20

✓ 0/1 pt ↻ 5

If $\{\vec{v}_1, \dots, \vec{v}_p\}$ are vectors in $\mathbb{R}^n$, then $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$ is a subspace of $\mathbb{R}^p$.

- ☐ True
☐ False

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Question 21

0/1 pt 5

Let $V = \{\vec{u}, \vec{v}, \vec{w}\}$ where $\vec{u}$, $\vec{v}$, and $\vec{w}$ are vectors in $\mathbb{R}^3$.

What are the possibilities for $\text{Span}(V)$?

- ☐ A line containing origin
- ☐ A line not containing origin
- ☐ A plane containing the origin
- ☐ The Origin
- ☐ A plane not containing origin
- ☐ Three dimensional space
- ☐ A point other than the origin

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Question 22

0/1 pt 5

Why is the set

$$\left\{ \begin{bmatrix} -91 \\ 13 \\ 0 \end{bmatrix}, \begin{bmatrix} 89 \\ 80 \\ 0 \end{bmatrix}, \begin{bmatrix} 156 \\ 217 \\ 0 \end{bmatrix} \right\}$$

not a basis for $\mathbb{R}^3$? Select all that apply.

- ☐ The set is linearly dependent.
- ☐ It is not a spanning set.
- ☐ The set contains too many elements.
- ☐ The set contains too few elements.
- ☐ It is not a subset of $\mathbb{R}^3$

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Question 23

0/1 pt 5

Identify each statement as true or false.

☐ If $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is a set of vectors in $\mathbb{R}^n$ with k less than n , then $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is linearly dependent.

☐ If the homogeneous linear system with augmented matrix $[\vec{v}_1 \dots \vec{v}_n, \vec{0}]$ has a nontrivial solution, the set $\{\vec{v}_1, \dots, \vec{v}_n\}$ can be linearly independent.

☐ If $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is a set of vectors in $\mathbb{R}^n$ with k greater than n , then $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is linearly dependent.

☐ If the homogeneous linear system with augmented matrix $[\vec{v}_1 \dots \vec{v}_n, \vec{0}]$ has a nontrivial solution, the set $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly dependent.

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● Question 24

0/1 pt 5

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ be vectors. Which of the following statements is **equivalent** to the statement "the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ are linearly independent"?

- ☐ The equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$ has no solutions.
- ☐ The equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$ has a unique solution.
- ☐ The equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$ has infinitely many solutions.

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● Question 25

0/1 pt 5

Which of the following best characterizes when a set of one vector $\{\mathbf{v}\}$ is linearly **independent**?

- ☐ When $\mathbf{v}$ is a multiple of another vector.
- ☐ When $\mathbf{v}$ is the zero vector.
- ☐ When $\mathbf{v}$ has more entries than columns.
- ☐ When $\mathbf{v}$ is any nonzero vector.

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● Question 26

0/1 pt 5

Which of the following conditions would imply that a set of two vectors $\{\mathbf{u}, \mathbf{v}\}$ is linearly **dependent**?

- ☐ $\mathbf{u}$ and $\mathbf{v}$ have the same number of entries.
- ☐ $\mathbf{u}$ is a scalar multiple of $\mathbf{v}$.
- ☐ $\mathbf{u}$ and $\mathbf{v}$ have a different number of entries.
- ☐ $\mathbf{u}$ and $\mathbf{v}$ are nonzero and point in different directions.
- ☐ $\mathbf{u}$ is the zero vector.

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● Question 27

✓ 0/1 pt ↻ 5

Which of the following is a **necessary** condition for a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ to be linearly **dependent**?

In other words, if all you know is that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is a linearly dependent set, which of the following conditions **must** be true?

- ☐ One of the vectors is a multiple of one of the other vectors.
- ☐ The equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$ has a nontrivial solution.
- ☐ One of the vectors is the zero vector.
- ☐ One of the vectors is a linear combination of the other vectors.
- ☐ $p > n$ ('there are more vectors than entries')

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● Question 28

✓ 0/1 pt ↻ 5

Which of the following conditions would imply that a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is linearly **dependent**?

- ☐ One of the vectors is the zero vector.
- ☐ $p > n$ ('there are more vectors than entries')
- ☐ One of the vectors is a linear combination of the other vectors.
- ☐ The equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0}$ has a nontrivial solution.
- ☐ One of the vectors is a multiple of one of the other vectors.

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● Question 29

✓ 0/1 pt ↻ 5

Are the vectors $\begin{bmatrix} 2 \\ -3 \\ -4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ -7 \\ -9 \end{bmatrix}$ linearly independent or linearly dependent?

- ☐ linearly independent
☐ linearly dependent

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● Question 30

0/1 pt 5

Are the vectors below linearly independent or linearly dependent?

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

- ☐ linearly dependent
☐ linearly independent

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● Question 31

0/1 pt 5

Determine if the set of polynomials is linearly independent or linearly dependent.

$$\{ 3x^2 - x + 6, \quad 30x^2 - 46x + 84, \quad 6x - 4 \}$$

These polynomials are linearly

- ☐ linearly dependent
☐ linearly independent

Show your work (a screenshot of octave is sufficient.)

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● Question 32

0/1 pt 5

For the vectors $\vec{u} = \begin{bmatrix} 2 \\ -4 \\ -3 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 4 \\ -12 \\ 3 \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} 2 \\ -12 \\ 15 \end{bmatrix}$, you notice that $\vec{w} = -3\vec{u} + 2\vec{v}$.

Is the set $\{\vec{u}, \vec{v}, \vec{w}\}$ linearly independent, linearly dependent, or is this not possible to determine?

Answer:

- ☐ linearly dependent
- ☐ impossible to tell
- ☐ linearly independent

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● Question 33

✓ 0/1 pt ↻ 5

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ -4 \\ -3 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} -2 \\ 4 \\ -4 \end{bmatrix}$.

Is the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent?

- ☐ Yes, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.
- ☐ No, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent.

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● Question 34

✓ 0/1 pt ↻ 5

For each subset of the xy -plane, decide if it is a subspace of $\mathbb{R}^2$.

- The set of points where $x = 0$.
 - ☐ Subspace of $\mathbb{R}^2$
 - ☐ Not a subspace of $\mathbb{R}^2$
- The set of points where $x \geq 0$.
 - ☐ Subspace of $\mathbb{R}^2$
 - ☐ Not a subspace of $\mathbb{R}^2$
- The set of points where $y = 0$.
 - ☐ Subspace of $\mathbb{R}^2$
 - ☐ Not a subspace of $\mathbb{R}^2$
- The set of points where $x = -5$.
 - ☐ Subspace of $\mathbb{R}^2$
 - ☐ Not a subspace of $\mathbb{R}^2$
- The set of points where $y = 4$.
 - ☐ Subspace of $\mathbb{R}^2$
 - ☐ Not a subspace of $\mathbb{R}^2$

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● Question 35

✓ 0/1 pt ↻ 5

Select the subset(s) that are subspaces.

- ☐ The set of all vectors in $\mathbb{R}^2$ of the form $x_1 \begin{bmatrix} 1 \\ 7 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ 1 \end{bmatrix}$ where x_1, x_2 are real numbers
- ☐ The set of all vectors in $\mathbb{R}^2$ whose endpoint lies on the line $y = 7x - 1$.
- ☐ The set of all 2×2 matrices of the form $\begin{bmatrix} a & 3 \\ b & 0 \end{bmatrix}$ where $a, b \in \mathbb{R}$.
- ☐ All polynomials in P_3 that have no t^2 term.

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● Question 36

✓ 0/1 pt ↻ 5

For each subset of $\mathbb{R}^2$, tell if it is a subspace of $\mathbb{R}^2$:

- the set of points that lie on the x -axis
- the set of points that lie in the third quadrant
- The set of points $\{(x, 5) : x \text{ is a real number}\}$

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● Question 37

✓ 0/1 pt ↻ 5

Which of the following is **not** a subspace of $\mathbb{R}^2$?

- ☐ The set $\left\{ \begin{bmatrix} 2 \\ -3 \end{bmatrix} \right\}$.
- ☐ The set $\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$.
- ☐ The set containing the points on the graph of the line $y = 2x - 3$.
- ☐ The set containing the points on the graph of the line $y = -\frac{2}{3}x$.
- ☐ The set $\mathbb{R}^2$.
- ☐ The set containing all multiples of $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

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● Question 38

✓ 0/1 pt ↻ 5

Part 1 of 2

$$\text{Let } H = \text{span} \left\{ \begin{bmatrix} -1 \\ 6 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 8 \\ 0 \\ -4 \end{bmatrix}, \begin{bmatrix} 3 \\ -18 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

Find the dimension of H .

● Question 39

✔ 0/1 pt ↻ 5

The set

$$\{ -4 \sin(\theta) + 6 \cos(\theta), -\sin(\theta) + 15 \cos(\theta), 3 \sin(\theta) + 9 \cos(\theta) \}$$

is a subset of $C(\mathbb{R})$ and contains a vector that is the sum of the other two. Use this information to find a proper subset with the same span.

- ☐ $\{ -4 \sin(\theta) + 6 \cos(\theta), -\sin(\theta) + 15 \cos(\theta), 3 \sin(\theta) + 9 \cos(\theta) \}$
- ☐ $\{ -\sin(\theta) + 15 \cos(\theta), 3 \sin(\theta) + 9 \cos(\theta) \}$
- ☐ $\{ -4 \sin(\theta) + 6 \cos(\theta), 3 \sin(\theta) + 9 \cos(\theta) \}$
- ☐ $\{ -4 \sin(\theta) + 6 \cos(\theta), -\sin(\theta) + 15 \cos(\theta) \}$
- ☐ $\{ -4 \sin(\theta) + 6 \cos(\theta) \}$
- ☐ $\{ -\sin(\theta) + 15 \cos(\theta) \}$
- ☐ $\{ 3 \sin(\theta) + 9 \cos(\theta) \}$
- ☐ $\{ \}$

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● Question 40

✔ 0/1 pt ↻ 5

Let W be a set of vectors in $\mathbb{R}^n$. According to the definition, which one of the following is NOT a property of W being a subspace?

- ☐ $\vec{u} \cdot \vec{v}$ is in W whenever $\vec{u}$ and $\vec{v}$ are vectors in W .
- ☐ The zero vector of $\mathbb{R}^n$ is contained in W .
- ☐ $k\vec{u}$ is in W whenever $\vec{u}$ is a vector in W and k is a scalar.

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● Question 41

✔ 0/1 pt ↻ 5

Suppose H is a linear transformation from $\mathbb{R}^3$ to $M_{2 \times 2}$, and suppose we know that

$$H\left(\begin{bmatrix} -5 \\ 3 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 3 & 2 \\ 5 & 6 \end{bmatrix} \text{ and } H\left(\begin{bmatrix} -4 \\ 0 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 4 & 2 \\ 3 & 9 \end{bmatrix}.$$

Use the fact that $\begin{bmatrix} -21 \\ 3 \\ 7 \end{bmatrix} = 4 \begin{bmatrix} -4 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} -5 \\ 3 \\ -1 \end{bmatrix}$ in order to find $H\left(\begin{bmatrix} -21 \\ 3 \\ 7 \end{bmatrix}\right)$.

Answer:

Hint

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● Question 42

✓ 0/1 pt ↻ 5

Suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation and that $\mathbf{u} \in \mathbb{R}^n$ is a vector such that $T(\mathbf{u}) = \mathbf{0}$.

Based on just this information, which of the following statements **must** be true?

- ☐ $T(2\mathbf{u}) = \mathbf{0}$.
- ☐ $T(\mathbf{x}) = \mathbf{0}$ is consistent.
- ☐ $T(\mathbf{x}) = \mathbf{0}$ has infinitely many solutions.
- ☐ $T(-\mathbf{u}) = \mathbf{0}$.
- ☐ $T(\mathbf{x}) = \mathbf{u}$ is consistent.
- ☐ $\mathbf{u}$ is the zero vector.

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● Question 43

✓ 0/1 pt ↻ 5

Find the standard matrix for the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates points about the origin by $\frac{\pi}{6}$ radians and then dilates points vertically by a factor of $\frac{17}{1}$.

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● Question 44

✓ 0/1 pt ↻ 5

Let T be a linear transformation induced by the matrix $A = \begin{bmatrix} -1 & -1 \\ 6 & 1 \end{bmatrix}$ and S a linear transformation induced by the matrix $B = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$. Find the matrix of $S \circ T$ and find $S \circ T(\vec{x})$ for $\vec{x} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

Matrix for $S \circ T$:

Matrix for $S \circ T(\vec{x})$:

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● Question 45

✓ 0/1 pt ↻ 5

Part 1 of 2

Let $L(x) = 5x$.

$L(x + y) =$

$L(x) + L(y) =$

$L(cx) =$

$cL(x) =$

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● Question 46

✓ 0/1 pt ↻ 5

Part 1 of 2

Let $L(x) = 3x - 7$.

$L(x + y) =$

$L(x) + L(y) =$

$L(cx) =$

$cL(x) =$

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● Question 47

✓ 0/1 pt ↻ 5

Let $T(a + bx + cx^2) = \begin{bmatrix} 0 \\ c \\ a \end{bmatrix}$ be a linear transformation from P_2 to $\mathbb{R}^3$.

Find a non-zero function in the null space of the transformation.

$p(x) =$

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● Question 48

✓ 0/1 pt ↺ 5

Is the linear transformation $F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 4 & 20 \\ 0 & 0 \\ 6 & 30 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

a. one-to-one? ? ▾

b. onto? ? ▾

c. an isomorphism? ? ▾

Question Help: [✉ Message instructor](#)

● Question 49

✓ 0/1 pt ↺ 5

Is the linear transformation $P : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $P\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 0 & 6 & 8 \\ 0 & -3 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

a. one-to-one? ? ▾

b. onto? ? ▾

c. an isomorphism? ? ▾

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● Question 50

✓ 0/1 pt ↺ 5

Is the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 7 & -5 \\ -5 & -3 \\ 6 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

a. one-to-one? ? ▾

b. onto? ? ▾

c. an isomorphism? ? ▾

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Question 51

0/1 pt 5

Is the linear transformation $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 11 \\ -4 & 11 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

a. one-to-one? ? ▾

b. onto? ? ▾

c. an isomorphism? ? ▾

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Question 52

0/1 pt 5

For which matrices M is the linear transformation T_M an isomorphism?

Any...

- ☐ M with more columns than rows
- ☐ M with linearly independent columns
- ☐ M with more rows than columns
- ☐ square M
- ☐ invertible M
- ☐ M with linearly independent rows

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